Measuring the Legacies of State Coordination

Democratization, Sectoral Support Structures, and Heterogeneous Innovation in South Korea

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Initial methodological note for discussion
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1. Research question

This project asks why some sectors experience broader firm entry and innovation after democratization while others remain concentrated among legacy incumbents. I focus on South Korea's 1987 democratic transition as a common political shock and ask whether post-transition sectoral outcomes depended on organizational support structures built under the earlier regime.

The broader motivation is a sequencing question: democratization may not operate on a blank institutional slate. It may interact with pre-existing state capacity, policy networks, and industrial coordination mechanisms. The empirical challenge is to measure those inherited structures more directly than conventional expenditure proxies allow.

2. Competing mechanisms

Capacity-building mechanism. Earlier industrial policies may have created durable coordination networks linking ministries, firms, research institutes, banks, and technology agencies. Under this mechanism, democratization may broaden access to these structures, reducing coordination frictions and supporting entry, experimentation, and innovation diffusion.

Incumbent-entrenchment mechanism. Alternatively, earlier policies may have created concentrated advantages for established firms. Under this mechanism, political opening does not necessarily broaden access; it may preserve legacy advantages, keeping innovation and market power clustered among incumbents.

The empirical goal is therefore not simply to ask whether democratization raised innovation on average, but whether inherited state coordination changed the distribution of post-democratization opportunities across sectors and firms.

3. Measurement: State Support Index

The key treatment variable is a sector-level measure of pre-1987 state support. Rather than relying only on spending levels or broad policy categories, I propose using archival materials - laws, industrial plans, R&D mandates, policy directives, and sectoral development documents - to construct a source-traceable **State Support Index**.

Let s index sectors, t index years, and $d \in \mathcal{D}$ index coding dimensions. The proposed dimensions are persistence, specificity, network breadth, and allocation. For a given sector-year-dimension cell, let C_{std} denote the coded score derived from archival passages. Define the pre-transition average for sector s on dimension d as:

$$\bar{C}_{sd} = \frac{1}{|\mathcal{T}_0|} \sum_{t \in \mathcal{T}_0} C_{std}, \qquad \mathcal{T}_0 = \{t : t < 1987\}.$$

A standardized composite index can then be written as:

$$S_s = \sum_{d \in \mathcal{D}} w_d \, \widetilde{C}_{sd}, \qquad \widetilde{C}_{sd} = \frac{\bar{C}_{sd} - \mu_d}{\sigma_d},$$

where w_d are pre-specified dimension weights and μ_d, σ_d are the mean and standard deviation of $\bar{C}_{sd}$ across sectors. In the initial analysis, I would report both the composite index S_s and dimension-specific results, since the allocation dimension may help distinguish broad coordination from incumbent-directed support.

This index should be interpreted as a measure of pre-transition organizational support, not as a direct causal measure of state capacity. The value of the index is that each score can be linked back to documentary evidence and reviewed against a fixed coding rubric.

4. Empirical design

The main design is a continuous-treatment event-study / generalized Difference-in-Differences framework. Let Y_{ust} denote an outcome for unit u in sector s at time t. Depending on the outcome, u may be a sector, industry, or firm. Outcomes may include firm entry, patenting, new-firm patenting, incumbent patenting, and concentration measures such as the Herfindahl-Hirschman Index (HHI).

Let $T^* = 1987$ be the transition year and define event-time indicators:

$$D_t^k = \mathbf{1}\{t - T^* = k\}.$$

The preferred dynamic specification is:

$$Y_{ust} = \alpha_u + \lambda_t + \sum_{k \neq -1} \beta_k \left(D_t^k \times \widetilde{S}_s \right) + \mathbf{Z}_{s0}' \boldsymbol{\delta}_t + \varepsilon_{ust}.$$

Here α_u are unit fixed effects, λ_t are year fixed effects, and $\widetilde{S}_s$ is the standardized pre-1987 State Support Index. The omitted event period is $k = -1$, so each β_k is interpreted relative to the year immediately before transition. The vector $\mathbf{Z}_{s0}$ includes baseline sector characteristics measured before democratization, such as capital intensity, export exposure, pre-transition concentration, or pre-transition innovation levels, interacted flexibly with year effects through $\boldsymbol{\delta}_t$.

Because $\widetilde{S}_s$ is fixed before 1987, its main effect is absorbed by unit fixed effects. The year fixed effects absorb shocks common to all sectors. The coefficients β_k therefore trace whether high-support and low-support sectors diverged around the democratic transition. Pre-transition coefficients provide a diagnostic for differential pre-trends:

$$H_0 : \beta_k = 0 \quad \text{for all } k < 0.$$

A more compact post-period specification can summarize average post-transition divergence:

$$Y_{ust} = \alpha_u + \lambda_t + \theta \left(\text{Post}_t \times \widetilde{S}_s \right) + \mathbf{Z}_{s0}' \boldsymbol{\delta}_t + \varepsilon_{ust},$$

where $\text{Post}_t = \mathbf{1}\{t \geq 1987\}$. This specification is less informative than the event-study design but useful for summary tables and robustness checks.

5. Mechanism interpretation

The mechanisms imply different patterns across outcomes. Under the capacity-building mechanism, higher pre-transition support should be associated with broader post-transition entry and innovation diffusion:

$$\beta_k^{\text{entry}} > 0, \qquad \beta_k^{\text{new-firm patents}} > 0, \qquad \beta_k^{\text{HHI}} < 0 \quad (k \geq 0).$$

Under the incumbent-entrenchment mechanism, higher pre-transition support should be associated with persistent concentration or incumbent-focused innovation:

$$\beta_k^{\text{entry}} \leq 0, \qquad \beta_k^{\text{incumbent patents}} > 0, \qquad \beta_k^{\text{HHI}} > 0 \quad (k \geq 0).$$

In addition to the composite index, dimension-specific specifications can examine whether different forms of support have different implications:

$$Y_{ust} = \alpha_u + \lambda_t + \sum_{k \neq -1} \sum_{d \in \mathcal{D}} \beta_{kd} \left(D_t^k \times \widetilde{C}_{sd} \right) + \mathbf{Z}'_{s0} \boldsymbol{\delta}_t + \varepsilon_{ust}.$$

This helps test whether persistence, specificity, network breadth, or allocation patterns are driving the observed relationship.

6. Identification concerns and robustness

The central concern is that pre-democratic support may reflect the regime's selection of already-promising sectors rather than the creation of reusable state capacity. I would address this in four ways.

First, the event-study coefficients before 1987 are used to inspect whether high-support and low-support sectors were already diverging. Second, baseline sector characteristics are interacted with year effects to account for differential exposure to capital intensity, export orientation, pre-transition innovation, or initial concentration. Third, I would test sensitivity to global demand shocks and export exposure. Fourth, I would compare the archival State Support Index to simpler measures such as expenditure proxies or broad policy categories.

Additional robustness checks would include alternative weights for the index, leave-one-dimension-out tests, placebo transition dates, and separate analyses of entry, patenting, and concentration. Standard errors would be clustered at the sector level where treatment varies; if the number of sectors is small, I would consider wild-cluster bootstrap or randomization-inference style checks.

The design should be interpreted as a differential-response design around a common political transition. Its credibility depends on whether the pre-transition support index is fixed before democratization, whether pre-trends are limited, and whether the archival coding process is transparent enough to be audited.

7. AI-assisted measurement workflow

The measurement workflow is intended to be constrained, auditable, and reproducible. Historical passages would be retrieved and coded using a fixed rubric, structured output fields, and source-level documentation. The goal is not to replace human judgment, but to scale a transparent coding process while preserving a reviewable audit trail.

Validation would proceed through hand-coded benchmark samples, prompt/output logging, inter-coder checks where possible, and systematic error analysis. Ambiguous cases would be routed to human review. This structure is intended to mitigate hallucination risk and make each score traceable to documentary evidence.

Near-term outputs would include: (i) a sector-policy corpus with document metadata, (ii) a hand-coded benchmark sample, (iii) a validation memo comparing machine-assisted scores with benchmark labels, (iv) a cleaned sector-year panel, and (v) initial event-study figures and robustness tables.

8. Expected contribution

Substantively, the project clarifies how democratization interacts with inherited industrial organization and state capacity. Methodologically, it develops a transparent approach for turning historical policy text into auditable, sector-level measures that can be used in panel-data designs.

For an AI-native research environment, the project is also useful as a workflow problem: how to increase research efficiency while keeping measurement, validation, and causal interpretation under human control.